

# Lokesh Venkatesh

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Research Interests: **Complex Systems** | **Computational Neuroscience** | **Machine Learning**



## EDUCATION

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| <b>Indian Institute of Science Education and Research (IISER), Pune</b><br><i>Integrated BS-MS   Natural Sciences — Interdisciplinary coursework across the basic sciences</i> | Nov 2022 — Present<br>CGPA: 7.7/10 |
| <b>Yuvabharathi Public School, Coimbatore</b><br><i>AISSCE   Grade 12 (CBSE) — Physics, Mathematics, Chemistry, Informatics Practices, English</i>                             | Jul 2022<br>GRADE: 92.6%           |
| <b>Yuvabharathi Public School, Coimbatore</b><br><i>AISSE   Grade 10 (CBSE) — Science, Mathematics</i>                                                                         | Jul 2020<br>GRADE: 94.4%           |

## RESEARCH EXPERIENCE

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| <b>Tumour-immune microenvironment interactions in cancer progression</b><br><i>Advisors: Prof. Ram Rup Sarkar, NCL Pune &amp; Dr. Chandrakala Meena, IISER Pune</i>                                                                                                                                                                                                                                                                                           | Nov 2025 — Present  |
| <ul style="list-style-type: none"> <li>Carrying out fixed-time simulations of an ODE model with further parameter &amp; bifurcation analyses to investigate how tumour-immune microenvironment interactions influence the dynamics of cancer progression.</li> </ul>                                                                                                                                                                                          |                     |
| <b>Self-supervised Representation Learning for Circadian Time Series</b><br><i>Advisor: Dr. Shaon Chakrabarti, NCBS Bangalore</i>                                                                                                                                                                                                                                                                                                                             | May 2025 — Jul 2025 |
| <ul style="list-style-type: none"> <li>Implemented representation learning algorithms like the variational autoencoder architecture on patient actigraphy datasets, followed by clustering, in order to obtain their learned low-dimensional representations.</li> <li>Analysed the learnt embeddings to identify the respective circadian phases and isolate what specific variables carry the most significance in the multivariate time series.</li> </ul> |                     |
| <b>Study of the Mitochondrial Genome using Motif Distributions</b><br><i>Advisor: Prof. MS Madhusudhan, IISER Pune</i>                                                                                                                                                                                                                                                                                                                                        | Aug 2024 — Apr 2025 |
| <ul style="list-style-type: none"> <li>Analysed motif distributions across a database of mitochondrial genomes as a metric to both reconstruct phylogenetic relationships over the course of its evolution, as well as identify genomic translocations.</li> <li>Also looked at possible regulatory correlations between mitochondrial &amp; nuclear genome promoter sequences.</li> </ul>                                                                    |                     |

## ACADEMIC PROJECTS & TRAINING

### SCHOOLS & WORKSHOPS

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| <b>Spins, Games &amp; Networks, 2025</b><br>Two-week fully-funded winter school on theoretical & computational approaches to complexity science at IMSc Chennai   | Dec 2025 |
| <b>NeuroAI</b><br>Ten-day interactive online course on principles connecting artificial & biological neural nets by the Neuromatch Academy                        | Jul 2025 |
| <b>Brain, Dynamics &amp; Computation, 2025</b><br>Two-week fully-funded summer school on network science-approaches to computational neuroscience at IMSc Chennai | May 2025 |

### COURSE & READING PROJECTS

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| <b>Tools for Complexity Science — Reading Project</b><br>Undertaking a reading project ( <i>Advisor: Dr. Chandrakala Meena</i> ) to study the various methods & tools, largely based on Stefan Thurner et al's book, used to model complex biological and social systems' phenomena. | Sep 2025 — Present |
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## RELEVANT COURSEWORK

**Data Science** — Applied Mathematical Methods, Deep Learning, Image & Video Processing, Mathematical & Statistical Optimisation, Computational & Functional Genomics, Signal Processing, Mathematical Finance

**Physics** — Classical Mechanics, Statistical Mechanics, Nonlinear Dynamics, Physics of Biological Systems

**Mathematics** — Multivariable Calculus, Linear Algebra, Probability, Graph Theory, Markov Processes

**Biology** — Systems Biology, Neurobiology, Bioinformatics, Cellular Biophysics I & II, Animal Behaviour, Ecology & Evolution, Cell Biology, Physiology, Genetics

**Humanities & Social Sciences** — Science & the World, Philosophy of Science, Cognitive Bases of Science, Science & Society, Public Policy I & II

## SKILLS

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**Computational & Modeling:** Numerical simulation of ODE systems, Parameter exploration and bifurcation analysis, Time-series analysis, Representation learning algorithms, Bioinformatics Tools

**Scientific Computing:** High-performance computing, Large-scale data processing, Pipeline development

**Programming:** Python (NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Biopython), Julia, Bash, Markdown,  $\text{\LaTeX}$

**Languages:** English, Tamil (bilingual proficiency); Hindi (working proficiency)

## SCHOLARSHIPS & AWARDS

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### INSPIRE's Scholarship for Higher Education (SHE)

INSPIRE-SHE Scholar (4 Lakh INR  $\approx$  4,500 USD) - awarded to top 0.1% (approx. 10,000) students nationwide by the Department of Science and Technology, Government of India

## EXTRACURRICULAR & OUTREACH

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### Academic Support Group, IISER Pune

Aug 2023 — Sep 2024

Senior teaching & logistics volunteer for a student collective mentoring socio-economically underprivileged students

### Kalpa — IISER Pune's Student Media House

Mar 2023 — Jan 2024

Member of the student media body's core team that coordinated investigative journalism projects on campus matters

## REFERENCES

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**Prof. M.S. Madhusudhan (IISER Pune)**

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**Dr. Chandrakala Meena (IISER Pune)**

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**Dr. Shaon Chakrabarti (NCBS Bangalore)**

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